

Name: SOLUTIONS.

Total Time: 30 Minutes

Marks \_\_\_\_\_/24

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Resources Allowed – 1 A4 Page (1 Side) of Notes, Calculator

## SIMPLE INTEREST

## Question 1

1 marks

How much interest is earned on a Principal of \$4000, invested at a rate of 6% per annum for a term of 5 years? Circle the correct answer:

A \$1495

B \$1200 ✓

C \$1500

D \$1800

## Question 2

2 marks

A washing machine costs \$700. If it is paid off over three years at 15% per annum simple interest, how much interest is paid?

$$\begin{aligned} S.I. &= P \times R \times T \\ &= 700 \times 0.15 \times 3 \quad \checkmark \\ &= \$315 \text{ in interest paid} \quad \checkmark \end{aligned}$$

$$\begin{aligned} P &= 700 \\ R &= 0.15 \\ T &= 3 \end{aligned}$$

## Question 3

3 marks

Hetty invests \$500 at 6% per year for 10 months. Calculate the gross amount of her investment after 10 months?

$$\begin{aligned} S.I. &= P \times R \times T \\ &= 500 \times 0.06 \times \frac{10}{12} \quad \checkmark \\ &= \$25 \text{ interest} \quad \checkmark \end{aligned}$$

$$\begin{aligned} P &= 500 \\ R &= 0.06 \\ T &= \frac{10}{12} \end{aligned}$$

$$\begin{aligned} \text{Gross} &= 500 + 25 \quad \checkmark \\ &= \$525 \end{aligned}$$

## Question 4

2 marks

If a person invests \$400 and it earns \$300 simple interest in 6 years, calculate the interest rate per year.

$$S.I. = P \times R \times T$$

$$\frac{S.I.}{P \times T} = R$$

$$\frac{300}{400 \times 6} = R \quad \left(\frac{1}{2}\right)$$

$$0.125 = R \quad \left(\frac{1}{2}\right)$$

$$\begin{aligned} R &= 0.125 \times 100 \quad \left(\frac{1}{2}\right) \\ &= \underline{\underline{12.5\%}} \quad \left(\frac{1}{2}\right) \end{aligned}$$

$$\begin{aligned} P &= 400 \\ I &= 300 \\ T &= 6 \\ R &= ? \end{aligned}$$

# INDEX LAWS

## Question 5

1 mark

$$2^5 - 2^3 = \boxed{2^{5-3}} = 2^2 = 2 \times 2 = 4 \quad \checkmark$$

Which number makes the expression above correct? Circle the correct answer.

☒ A 4

B 8

C 24

D 28

## Question 6

4 marks

Simplify the following, leaving in index form.

a  $3^6 \times 3^2$

$$= 3^{6+2} = 3^8 \quad \checkmark$$

b  $8^7 \div 8^3$

$$= \frac{8^{7-3}}{1} = 8^4 \quad \checkmark$$

c  $(2^4)^5$

$$= 2^{4 \times 5} = 2^{20} \quad \checkmark$$

d  $\frac{5^7 \times 5^4}{5^6}$

$$= \frac{5^{7+4}}{5^6} = \frac{5^{11}}{5^6} = 5^{11-6} = 5^5 \quad \checkmark$$

## Question 7

2 marks

Evaluate the following

a  $3^{-5}$

$$\frac{1}{3^5} = \frac{1}{243} \quad \checkmark$$

b  $7t^0$

$$7 \times 1 = 7 \quad \checkmark$$

## Question 8

3 marks

Simplify the following

a  $s^5 \times s^7$

$$= s^{5+7} = s^{12} \quad \checkmark$$

b  $\frac{p^{12}}{p^3}$

$$= p^{12-3} = p^9 \quad \checkmark$$

c  $(f^7)^3$

$$= f^{7 \times 3} = f^{21} \quad \checkmark$$

## Question 9

6 marks

Simplify the following

a  $4e^2 \times 3e \times e^5$

$$= 12e^{2+1+5} = 12e^8 \quad \checkmark$$

b  $(x^3)^2 \times (3x^5)^3$

$$= x^{3 \times 2} \times 3^3 \times x^{5 \times 3} = x^6 \times 27x^{15} = 27x^{15+6} = 27x^{21} \quad \checkmark$$

c

$$\frac{a^6 b^{8-1}}{a^5 b} = a^{6-5} \times b^{8-1} = a \times b^7 = ab^7 \quad \checkmark$$